Mycelium Braille

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Raised dots grown from mycelium, to be read by eye and by finger.

Abstract

Can a living material work as a reading surface? To find out, I grow 10 × 10 cm plates of mycelium composite in 3D-printed molds whose cavities are domes. The mycelium fills them and becomes raised braille dots that write IFT, in three dot sizes, and I check each size by sight and by touch.

Introduction

Braille is read by touch, so a plate that carries it needs raised dots with a clear shape. Mycelium composite copies the surface it grows against, and the way it touches the mold changes its colour and texture [1]. The mold surface can therefore act as a writing tool, and the open question is whether it can form dots that are solid and clear enough to read.

Two things frame the project:

- The mold method from class (printed mold, lignocellulosic substrate, drying in the mold), which is my starting point.
- Mold contact and skin [1], which shows that what touches the mycelium shapes how it looks.

Scientific Questions

What we explore and why

Research question: can mycelium composite carry braille that can be read by eye and by touch?

Sub-questions:

1. Which dome size gives complete, solid dots?
2. Which mix of substrate gives domes that stay whole after demolding?
3. Can my classmates and I identify the letters IFT by sight, and separately by touch with our eyes closed?

Writing a short word is a concrete test: every dot has to come out whole and in the right place for the word to be read. It also asks mycelium to hold a message, not only a texture, in a form that a tactile data object could use [5].

State of the art

Braille in Art

Working with braille is a common thing in art and design world. The Blind, a French artist who uses braille language system to make grafitti in different country :



One of The Blind's work in Germany

Kotora Studio are making a lot of different art pieces with hand punched braille working on on material, surfaces and superposition of medium :



this very moment, 2025, KOTORA Studio

Ray Nachum, who did the cover for the album ANTI of Rihanna, puts braille in most of his work. In this one, the *INVISIBLE CHAIR* questions the meaning of vision itself. What does it mean to see ? How are we suppose to see ? This chair, "invisible" for most of us, holds a message for the one who know how to read differently, read with touch.



INVISIBLE CHAIR, Roy Nachum

Surface and contact

Contact between the mold and the mycelium changes the colour and pattern of the skin, and in the first trials of Globa et al. a fully continuous contact stopped the skin from forming [1]. The skin develops where substrate meets air, with denser hyphae than the interior [4]. My domes sit against the lid, so I watch whether the skin forms there.

Geometry and fidelity.

Geometry affects growth and strength in 3D-printed mycelium composites [2]. In molded acoustic tiles the mycelium copied the mold texture, with small irregularities in strongly curved areas, and in fully closed molds the humidity stayed high and the surface did not fully harden [3]. Both points apply to my domes and to my closed mold.

nformation as shape.

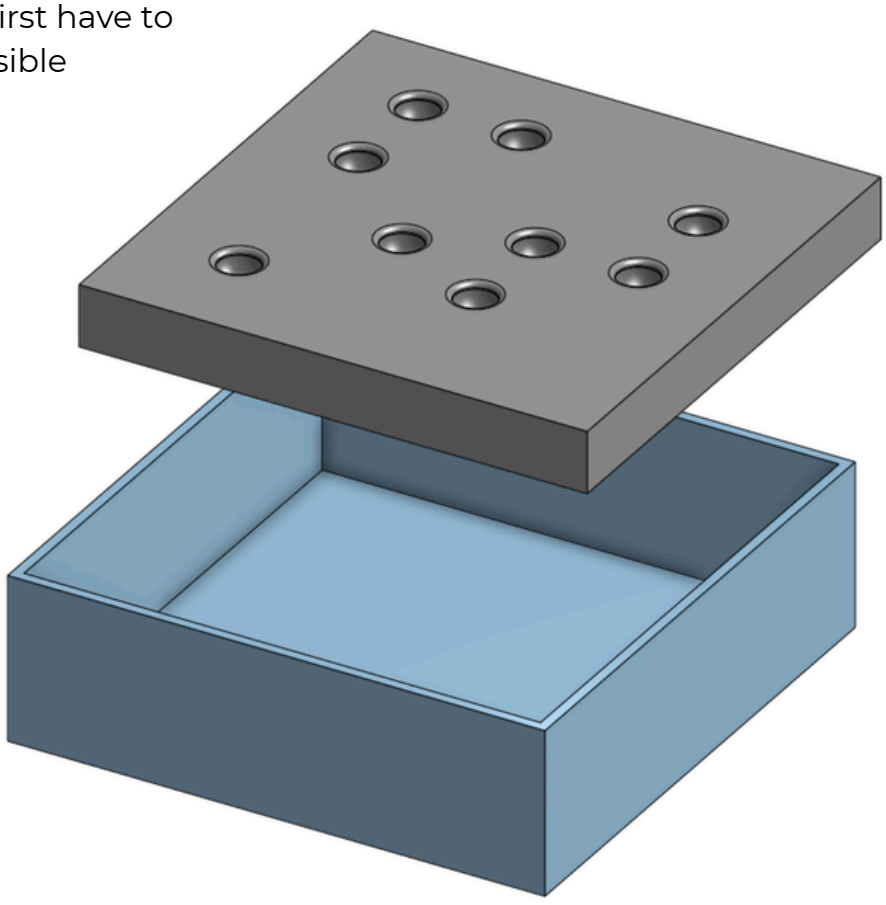
Tactile data physicalisation is mentioned in the literature as a theoretical application of myco-materials, highlighting their potential beyond conventional structural uses [5].

Mold design and concept

Three plates of 10 × 10 cm, one per dome size. The dots are half-ellipsoid domes defined by one number, the base diameter d. Each plate writes IFT in braille when the word fits and only IT or T when it doesn't fit.

Base diameter (d)	Height (0.66 x d)	Center spacing (d+d/4)	Letters on the plate
8mm	5.28	10 mm	IFT
10mm	6.66 mm	12.5 mm	IT
14mm	9.24 mm	17.5 mm	T

These dots are larger than standard braille. This is on purpose: before looking at smaller sizes, I first have to check that if it is even possible

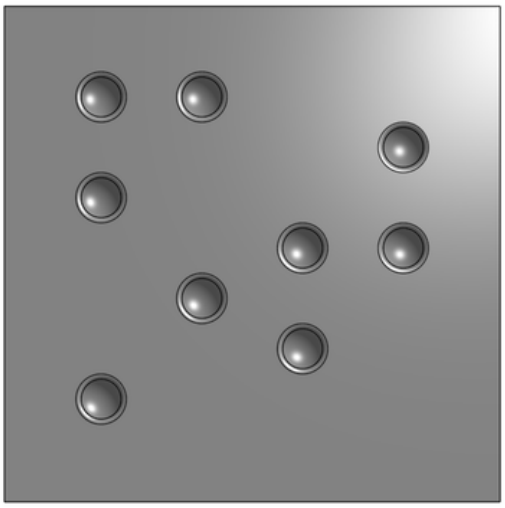


Mold design 3D models

Here are the molds :

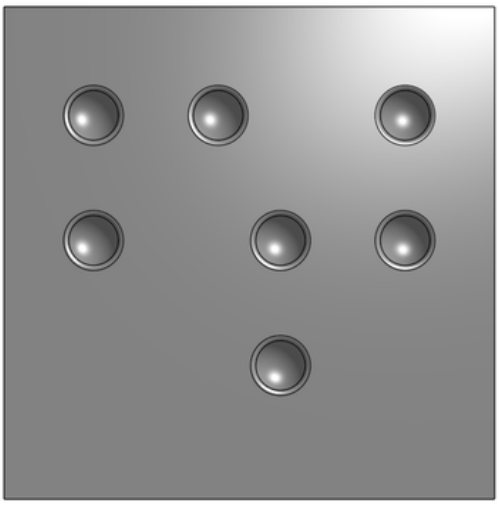
Three identical bottom parts and one cover for each dot diameter

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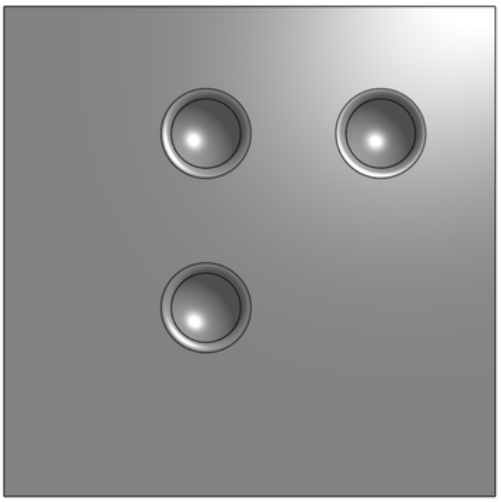
Top part of the mold with 8mm diameter dots.

I
F T



Top part of the mold with 10mm diameter dots.

I T



Top part of the mold with 14mm diameter dots.

I

References

1. Globa, A., Soh, E., & Le Ferrand, H. (2025). Living textures and mycelium skin co-creation: Designing colour, pattern, and performance for bio-aesthetic expression in mycelium-bound composites. Biomimetics, 10(9), 573. https://doi.org/10.3390/biomimetics10090573
2. Gomaa, H., Chau, W. M., Karazi, Y., Biala, E., Akbar, Z., Wortmann, T., & Ostermann, M. (2024). Influence of geometry on growth and strength of 3D-printed mycelium composites: A data-driven study. In Scalable Disruptors (pp. 331–345). Springer. https://doi.org/10.1007/978-3-031-68275-9_27
3. Mycelium-based composites for interior architecture: Digital fabrication of acoustic ceiling components (2025). Biomimetics, 10(11), 729 (authors to complete). https://doi.org/10.3390/biomimetics10110729
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5. Gough, P., Globa, A., Hadigheh, A., & Withana, A. (2022). Making sustainable, tangible objects with myco-materials [Tutorial]. ISS 2022. https://iss2022.acm.org/details/iss-2022-tutorials/2/Making-Sustainable-Tangible-Objects-with-Myco-materials